Page 2: The Basics – Electric Charge and Current

What is Electric Charge?

Charge is a fundamental property of matter, much like mass. It is responsible for all electric and magnetic phenomena.

- There are two types of charges: **Positive (+) and Negative (-)** .
- Like charges repel each other. Unlike charges attract each other.
- The SI unit of charge is the **Coulomb (C)** .

What is Electric Current?

Current is the **flow of electric charge**. To have a current, the charges need to move in a specific direction.

- **In metals (our usual conductors):** The charge carriers are **electrons**. They are negatively charged.
- **Convention:** By historical convention, we take the direction of flow of **positive charge** as the direction of current.
    - This means the conventional current flows from the **positive terminal** to the **negative terminal** of a battery.
    - The actual electron flow is from the negative terminal to the positive terminal. Don't let this confuse you! In your studies, always use the **conventional direction** (positive to negative) unless specified otherwise.

Mathematical Definition:

Current (I) is the rate of flow of charge.
If a charge 'Q' flows through a cross-section of a conductor in time 't', then:
I = Q / t

- The SI unit of current is the **Ampere (A)** .
- 1 Ampere = 1 Coulomb / 1 second.

Page 3: How Does Current Flow? – The Electric Circuit

For charge to flow continuously, it needs a closed path. This is called an **electric circuit**.

The "Water Analogy"

A very helpful way to visualize electricity is to compare it to a water flow system.

Electrical Quantity	Water Analogy
Electric Current (I)	**Flow Rate of Water** (liters per second)
Battery / Cell	**Water Pump** (provides pressure to push water)
Wire	**Pipe** (carries the water)
Resistor / Bulb	**A narrow section or a water wheel** (restricts flow, converts energy)
Switch	**A Tap / Valve** (starts or stops the flow)

Components of a Circuit:

1. **A Source of Energy:** A cell or battery that provides the "push" to the charges. This push is due to a chemical reaction that creates an excess of electrons at one terminal (-) and a deficit at the other (+).
2. **Conducting Wires:** Provide the path for the current to flow. Made of good conductors like copper or aluminum.
3. **A Load:** A device that uses electrical energy to do work (e.g., a bulb that gives light, a fan that rotates, a heater that produces heat).
4. **A Switch:** A device to complete (ON) or break (OFF) the circuit. Current flows only when the circuit is closed (switch is ON).

Page 4: The Electric "Push" – Potential Difference

What makes the charges flow? It's the same reason water flows – a difference in pressure or level.

Electric Potential
Just as a stone at a height has gravitational potential energy, a charge in an electric field has **electric potential energy**.

- **Electric Potential (V)** at a point is the amount of work done to bring a unit positive charge from infinity to that point.

Potential Difference
Charges flow from a point of higher electric potential to a point of lower electric potential. The difference in potential between two points is the driving force for the current.

Definition:
Potential Difference (V) between two points in a circuit is the amount of work done (W) in moving a unit charge (Q) from one point to the other.

V = W / Q

- The SI unit of potential difference is the **Volt (V)** .
- 1 Volt = 1 Joule / 1 Coulomb.
- If 1 joule of work is done to move 1 coulomb of charge between two points, the potential difference is 1 volt.

The Cell as a Pump:
A cell or battery does the work to lift the charge from a point of low potential (negative terminal) to a point of high potential (positive terminal), creating the potential difference that drives the current around the circuit.

Page 5: Opposition to Flow – Electric Resistance

As the charges flow through a conductor, they collide with the atoms (ions) of the material. This hinders their flow. This property of a conductor to oppose the flow of current is called **Resistance**.

Definition:
Resistance (R) is the opposition offered by a conductor to the flow of electric current.

- The SI unit of resistance is the **Ohm (Ω)** .
- A conductor has a resistance of 1 ohm if a current of 1 ampere flows through it when a potential difference of 1 volt is applied across its ends.

Cause of Resistance:
When electrons drift through a conductor, they constantly collide with the vibrating atoms (ions) of the metal lattice. These collisions transfer energy (which is why the wire heats up) and slow down the net flow of electrons.

Factors on which Resistance Depends:
The resistance (R) of a conductor depends on four factors:

1. **Length (l):** Resistance is directly proportional to the length of the wire. **($R \propto l$)** A longer wire offers more resistance.
2. **Area of Cross-Section (A):** Resistance is inversely proportional to the area of cross-section. **($R \propto 1/A$)** A thicker wire (larger area) offers less resistance.

3. **Nature of Material:** Different materials have different internal structures, so they offer different resistance. Copper has very low resistance, while rubber has extremely high resistance.
4. **Temperature (T):** For most metals, resistance increases with an increase in temperature.

Combining these, we get:
$R = \rho \times (l / A)$

Where **ρ (rho)** is a constant called **Resistivity** or **Specific Resistance**. It depends only on the nature of the material and the temperature, not on the dimensions of the wire.

Page 6: Ohm's Law – The Relationship

This is the most important law in the study of electricity. It relates the three quantities we've learned: Current (I), Potential Difference (V), and Resistance (R).

The Statement:
According to **Ohm's Law**, at a constant temperature, the current (I) flowing through a conductor is directly proportional to the potential difference (V) applied across its ends.

$V \propto I$

$V = I \times R$

Where R is a constant for the given conductor at a given temperature, known as its resistance.

Conceptual Understanding:

- Ohm's Law tells us that if you increase the voltage (the push), the current (the flow) will increase proportionally, provided the resistance (the opposition) stays the same.
- It also tells us that if you have a fixed voltage, a higher resistance will result in a lower current.

Ohmic vs. Non-Ohmic Conductors:

- **Ohmic Conductors:** Those that strictly obey Ohm's Law (V/I is constant). The V-I graph is a straight line passing through the origin. *Example: Metallic conductors like copper, aluminum at constant temperature.*
- **Non-Ohmic Conductors:** Those that do not obey Ohm's Law (V/I is not constant). Their resistance changes with voltage/current. *Example: Diode, LED, electrolyte, and even a filament bulb (because its resistance increases drastically as it gets hot).*

Page 7: Connecting Resistances – Series and Parallel

In a circuit, we often have more than one resistor. How they are connected drastically changes the overall (equivalent) resistance of the circuit.

1. Series Combination
Resistors are connected end-to-end, providing only one path for the current to flow.

- **Key Characteristic:** The **current is the same** through all resistors.
- **Voltage:** The total voltage (V) from the battery is divided (shared) among all the resistors. The sum of the individual voltage drops equals the total voltage.
    - $V = V_1 + V_2 + V_3$
- **Equivalent Resistance (R_s):** The total resistance offered by the combination is the sum of the individual resistances.
    - $R_s = R_1 + R_2 + R_3$
- **Effect:** The equivalent resistance is **greater than the largest individual resistance**. Connecting resistors in series increases the overall opposition to current.

2. Parallel Combination
Resistors are connected in such a way that the current is divided among multiple paths.

- **Key Characteristic:** The **voltage is the same** across all resistors (each is connected directly to the battery terminals).
- **Current:** The total current (I) from the battery is divided among the resistors. The sum of the individual currents equals the total current.
    - $I = I_1 + I_2 + I_3$
- **Equivalent Resistance (R_p):** For a parallel combination, the reciprocal of the total resistance is the sum of the reciprocals of the individual resistances.
    - $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$
- **Effect:** The equivalent resistance is **less than the smallest individual resistance** ($R_p < \text{smallest } R$). Connecting resistors in parallel provides more paths for the current, thus reducing the overall opposition.

Page 8: The Heating Effect of Electric Current

When current flows through a resistor (like the filament of a bulb or the coil of a heater), electrical energy is converted into heat energy. This is the **heating effect of electric current**.

Why does it happen?
As mentioned earlier, when electrons move through a conductor, they collide with the atoms/ions of the material. In these collisions, the kinetic energy lost by the electrons is transferred to the atoms, making them vibrate more vigorously. This increased vibration is what we perceive as **heat**.

What does the heat produced depend on?
The amount of heat (H) produced in a resistor depends on three factors:

1. **Current (I):** More current means more electrons, and thus more collisions per second.
2. **Resistance (R):** Higher resistance means more opposition, and thus more violent/energetic collisions.
3. **Time (t):** The longer the current flows, the more heat is generated.

Joule's Law of Heating:
The heat produced in a resistor is directly proportional to:

- The square of the current (I^2)
- The resistance (R)
- The time (t)

$$H = I^2 \times R \times t$$

Practical Applications:

1. **Electric Bulb:** A thin, high-resistance tungsten filament glows white-hot due to the heating effect, producing light. The bulb is filled with inert gas to prevent the filament from burning (oxidizing).
2. **Electric Iron / Heater / Toaster:** They use coils of a high-resistance alloy (like Nichrome) that become red hot and radiate heat.
3. **Electric Fuse:** This is a **safety device**. It is a short piece of wire made of a material with a low melting point (like tin-lead alloy). If the current exceeds a safe limit (due to a short circuit or overload), the fuse wire heats up, melts, and breaks the circuit, protecting expensive appliances from damage.

Page 9: Electric Power

What is Power?
In general, power is the rate of doing work. In electricity, **Electric Power** is the rate at which

electrical energy is consumed or converted into another form of energy (like heat, light, mechanical energy, etc.).

Derivation of Formula:
We know:

- Work done (Energy consumed) = Potential Difference × Charge moved = $V \times Q$
- Power (P) = Work done / Time = $(V \times Q) / t$
- But $Q / t = I$ (Current)

Therefore, we get the main formula for electric power:

$$\mathbf{P = V \times I}$$

- The SI unit of power is the **Watt (W)** .
- 1 Watt is the power consumed by a device if 1 Ampere of current flows through it when a potential difference of 1 Volt is applied across it.

Other Forms of the Power Formula:
Using Ohm's Law ($V = IR$), we can write the power formula in two other useful ways:

- $\mathbf{P = I^2 \times R}$ (Used when current and resistance are known)
- $\mathbf{P = V^2 / R}$ (Used when voltage and resistance are known)

Commercial Unit of Energy (kWh)
The joule is a very small unit for practical purposes. Electricity bills are based on a much larger unit called the **kilowatt-hour (kWh)** .

- **1 kilowatt-hour** is the amount of energy consumed by a device of power 1000 watts (1 kW) when it is used for 1 hour.
- **1 kWh = 1 unit** (as mentioned in your electricity bill).
- Relation between kWh and Joule:
    - $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$